



System Simulation Report

www.homerenergy.com

File: Remote_Facility_PV_Diesel_BEES.homer

Author: Rabih Al Ahmadih

Location: (0°0.0'N, 0°0.0'E)

Total Net Present Cost: \$166,159.40

Levelized Cost of Energy (\$/kWh): \$0.293

Notes: Techno-economic optimization of an off-grid PV-diesel-battery microgrid supplying a remote facility.

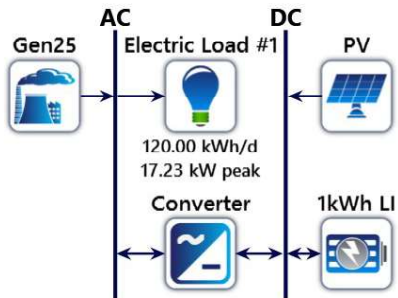
Sensitivity variable values for this simulation

Variable	Value	Unit
Diesel Fuel Price	1.20	\$/L
Solar Scaled Average	5.38	kWh/m ² /day

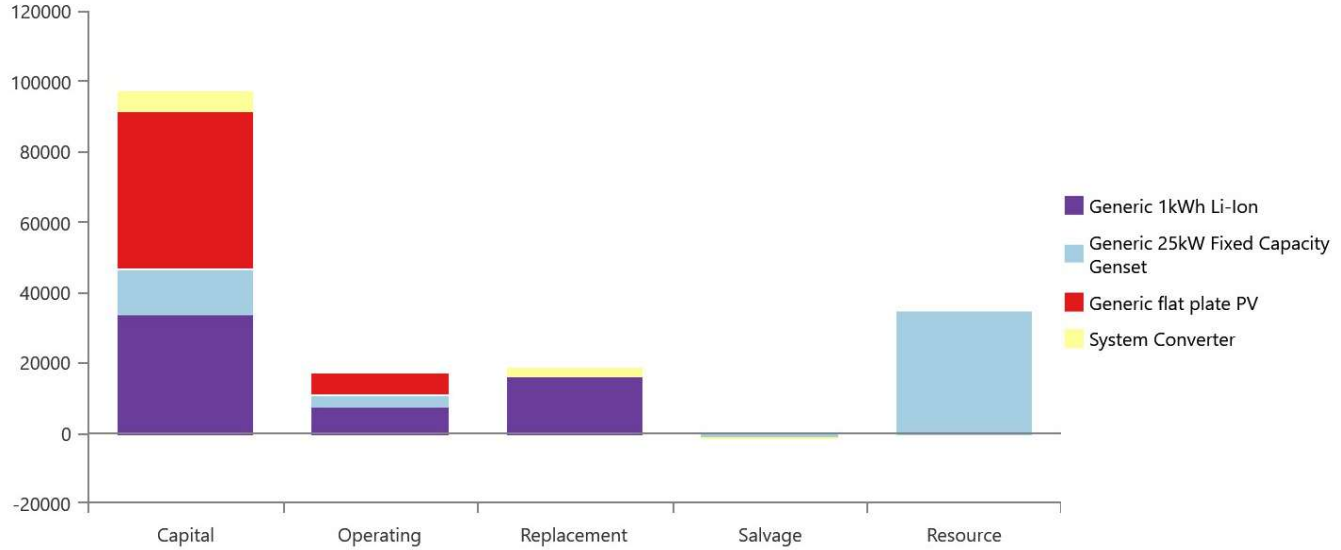
System Architecture

Component	Name	Size	Unit
Generator	Generic 25kW Fixed Capacity Genset	25.0	kW
PV	Generic flat plate PV	44.5	kW
Storage	Generic 1kWh Li-Ion	62	strings
System converter	System Converter	19.3	kW
Dispatch strategy	HOMER Cycle Charging		

Schematic



Cost Summary



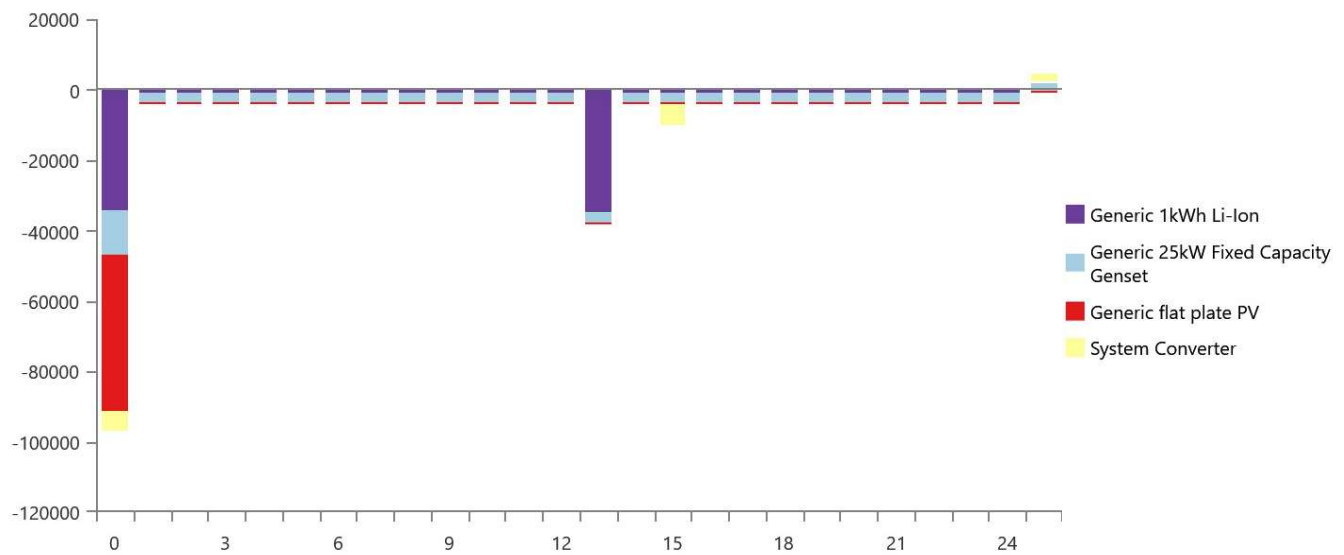
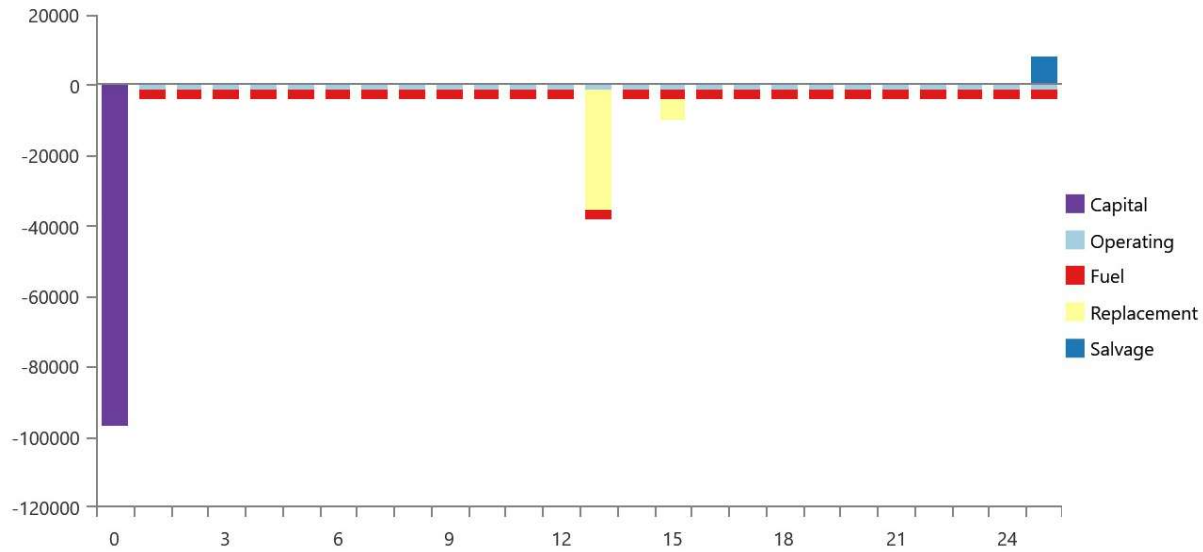
Net Present Costs

Name	Capital	Operating	Replacement	Salvage	Resource	Total
Generic 1kWh Li-Ion	\$34,100	\$8,015	\$16,572	-\$161.15	\$0.00	\$58,526
Generic 25kW Fixed Capacity Genset	\$12,500	\$3,238	\$0.00	-\$1,328	\$35,203	\$49,614
Generic flat plate PV	\$44,500	\$5,753	\$0.00	\$0.00	\$0.00	\$50,253
System Converter	\$5,777	\$0.00	\$2,451	-\$461.31	\$0.00	\$7,767
System	\$96,877	\$17,006	\$19,023	-\$1,950	\$35,203	\$166,159

Annualized Costs

Name	Capital	Operating	Replacement	Salvage	Resource	Total
Generic 1kWh Li-Ion	\$2,638	\$620.00	\$1,282	-\$12.47	\$0.00	\$4,527
Generic 25kW Fixed Capacity Genset	\$966.93	\$250.50	\$0.00	-\$102.69	\$2,723	\$3,838
Generic flat plate PV	\$3,442	\$445.00	\$0.00	\$0.00	\$0.00	\$3,887
System Converter	\$446.88	\$0.00	\$189.60	-\$35.68	\$0.00	\$600.80
System	\$7,494	\$1,316	\$1,471	-\$150.84	\$2,723	\$12,853

Cash Flow





Electrical Summary

Excess and Unmet

Quantity	Value	Units
Excess Electricity	29,446	kWh/yr
Unmet Electric Load	0	kWh/yr
Capacity Shortage	0	kWh/yr

Production Summary

Component	Production (kWh/yr)	Percent
Generic flat plate PV	69,929	90.5
Generic 25kW Fixed Capacity Genset	7,303	9.46
Total	77,232	100

Consumption Summary

Component	Consumption (kWh/yr)	Percent
AC Primary Load	43,800	100
DC Primary Load	0	0
Total	43,800	100

Generator: Generic 25kW Fixed Capacity Genset (Diesel)

Generic 25kW Fixed Capacity Genset Electrical Summary

Quantity	Value	Units
Electrical Production	7,303	kWh/yr
Mean Electrical Output	21.9	kW
Minimum Electrical Output	6.25	kW
Maximum Electrical Output	25.0	kW

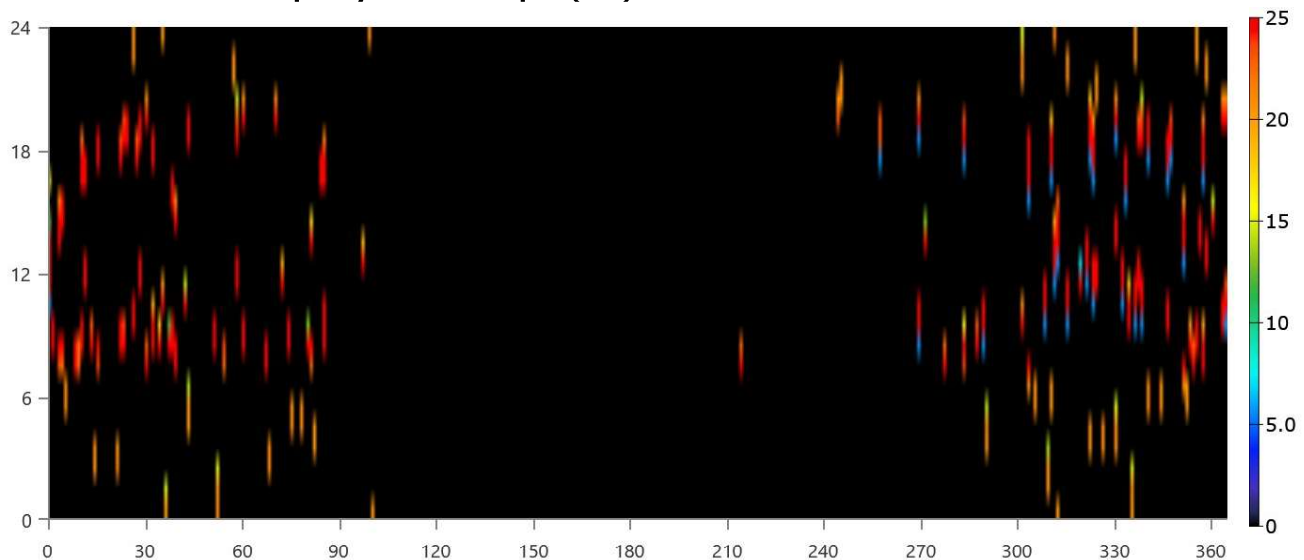
Generic 25kW Fixed Capacity Genset Fuel Summary

Quantity	Value	Units
Fuel Consumption	2,269	L
Specific Fuel Consumption	0.311	L/kWh
Fuel Energy Input	22,330	kWh/yr
Mean Electrical Efficiency	32.7	%

Generic 25kW Fixed Capacity Genset Statistics

Quantity	Value	Units
Hours of Operation	334	hrs/yr
Number of Starts	140	starts/yr
Operational Life	44.9	yr
Capacity Factor	3.33	%
Fixed Generation Cost	2.57	\$/hr
Marginal Generation Cost	0.328	\$/kWh

Generic 25kW Fixed Capacity Genset Output (kW)



PV: Generic flat plate PV

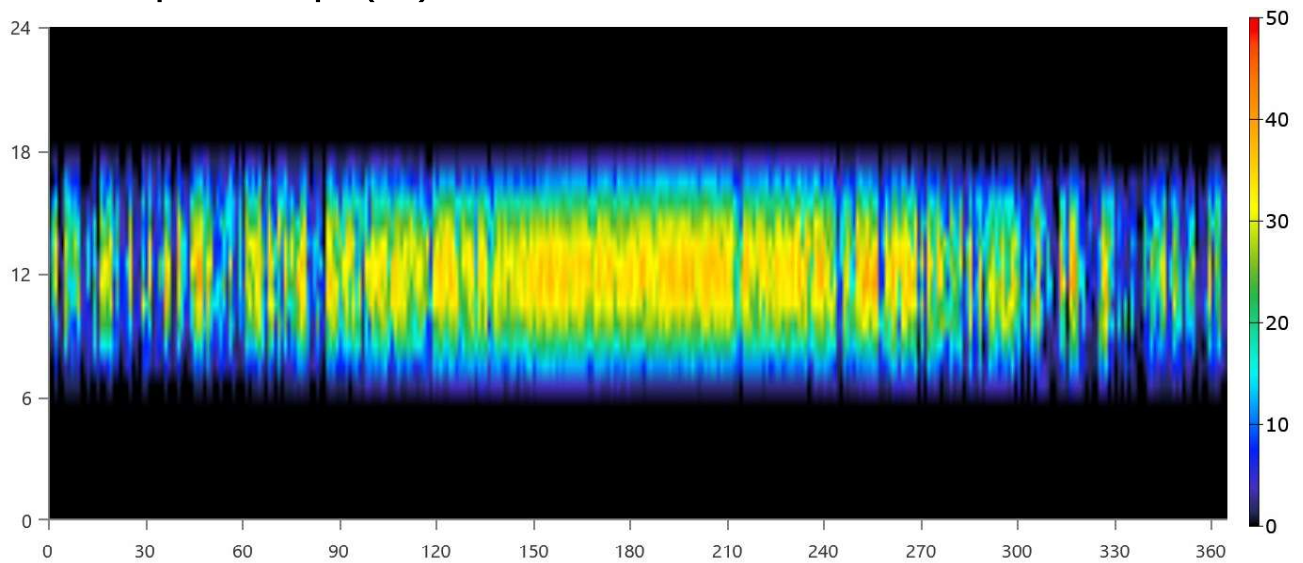
Generic flat plate PV Electrical Summary

Quantity	Value	Units
Minimum Output	0	kW
Maximum Output	42.1	kW
PV Penetration	160	%
Hours of Operation	4,380	hrs/yr
Levelized Cost	0.0556	\$/kWh

Generic flat plate PV Statistics

Quantity	Value	Units
Rated Capacity	44.5	kW
Mean Output	7.98	kW
Mean Output	192	kWh/d
Capacity Factor	17.9	%
Total Production	69,929	kWh/yr

Generic flat plate PV Output (kW)



Storage: Generic 1kWh Li-Ion

Generic 1kWh Li-Ion Properties

Quantity	Value	Units
Batteries	62.0	qty.
String Size	1.00	batteries
Strings in Parallel	62.0	strings
Bus Voltage	6.00	V

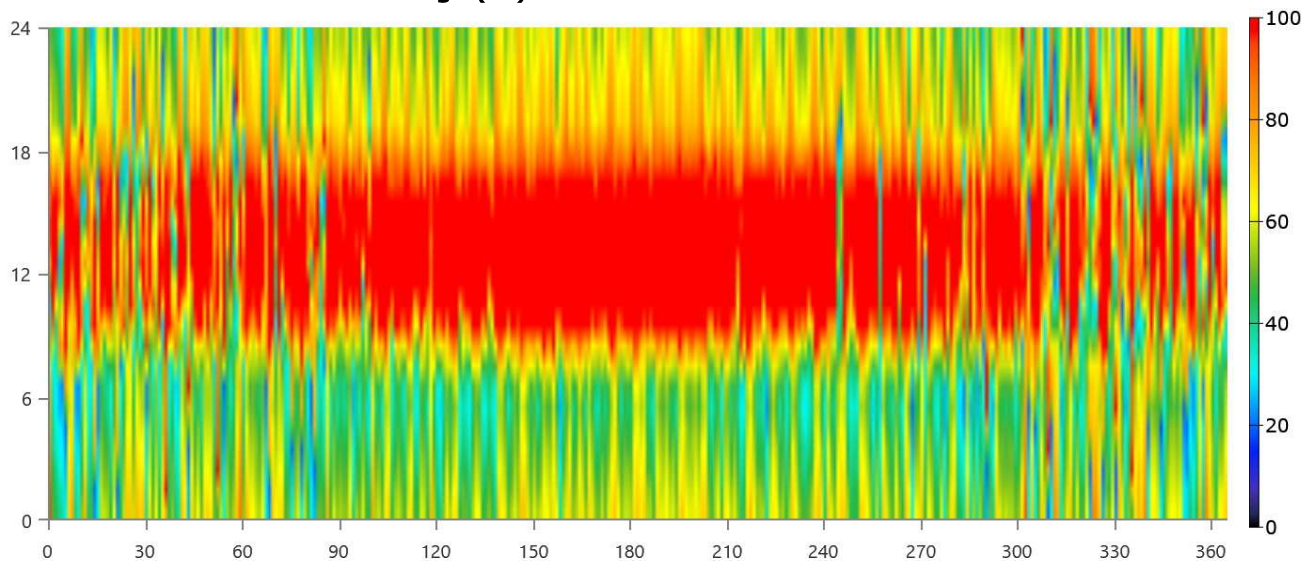
Generic 1kWh Li-Ion Result Data

Quantity	Value	Units
Average Energy Cost	0.144	\$/kWh
Energy In	15,513	kWh/yr
Energy Out	13,977	kWh/yr
Storage Depletion	16.4	kWh/yr
Losses	1,552	kWh/yr
Annual Throughput	14,733	kWh/yr

Generic 1kWh Li-Ion Statistics

Quantity	Value	Units
Autonomy	9.92	hr
Storage Wear Cost	0.193	\$/kWh
Nominal Capacity	62.0	kWh
Usable Nominal Capacity	49.6	kWh
Lifetime Throughput	186,000	kWh
Expected Life	12.6	yr

Generic 1kWh Li-Ion State of Charge (%)



Converter: System Converter

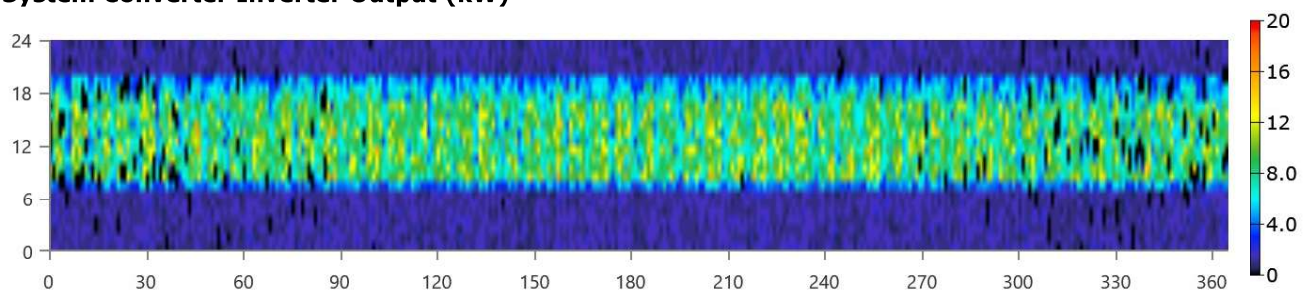
System Converter Electrical Summary

Quantity	Value	Units
Hours of Operation	8,455	hrs/yr
Energy Out	41,653	kWh/yr
Energy In	43,845	kWh/yr
Losses	2,192	kWh/yr

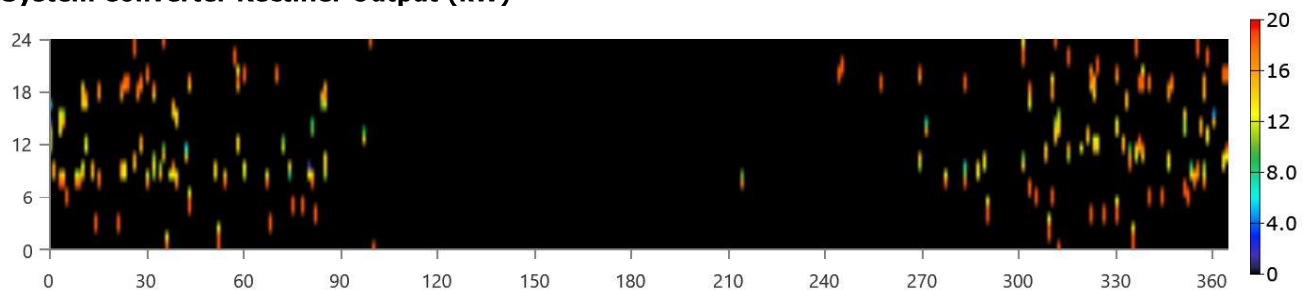
System Converter Statistics

Quantity	Value	Units
Capacity	19.3	kW
Mean Output	4.75	kW
Minimum Output	0	kW
Maximum Output	17.2	kW
Capacity Factor	24.7	%

System Converter Inverter Output (kW)



System Converter Rectifier Output (kW)

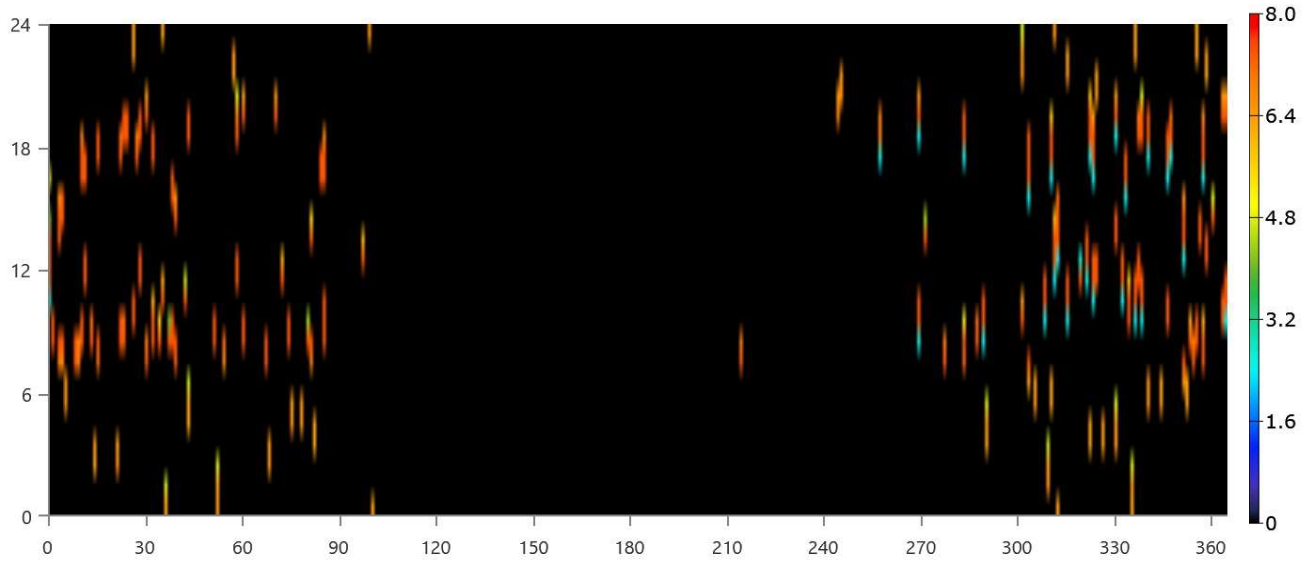


Fuel Summary

Diesel Consumption Statistics

Quantity	Value	Units
Total fuel consumed	2,269	L
Avg fuel per day	6.22	L/day
Avg fuel per hour	0.259	L/hour

Diesel Consumption (L/hr)



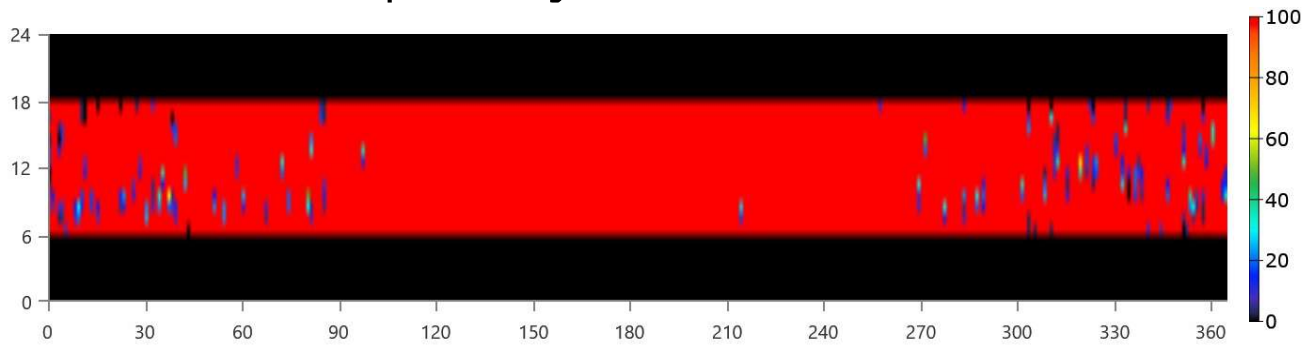
Emissions

Pollutant	Quantity	Unit
Carbon Dioxide	5,941	kg/yr
Carbon Monoxide	37.1	kg/yr
Unburned Hydrocarbons	1.63	kg/yr
Particulate Matter	0.222	kg/yr
Sulfur Dioxide	14.5	kg/yr
Nitrogen Oxides	34.9	kg/yr

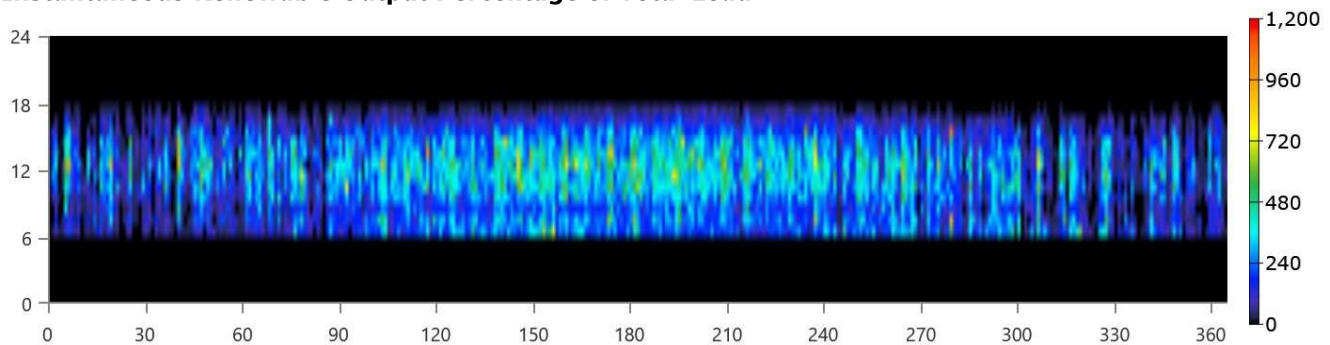
Renewable Summary

Capacity-based metrics	Value	Unit
Nominal renewable capacity divided by total nominal capacity	64.0	%
Usable renewable capacity divided by total capacity	58.7	%
Energy-based metrics	Value	Unit
Total renewable production divided by load	160	%
Total renewable production divided by generation	90.5	%
One minus total nonrenewable production divided by load	83.3	%
Peak values	Value	Unit
Renewable output divided by load (HOMER standard)	1,089	%
Renewable output divided by total generation	100	%
One minus nonrenewable output divided by total load	100	%

Instantaneous Renewable Output Percentage of Total Generation



Instantaneous Renewable Output Percentage of Total Load



100% Minus Instantaneous Nonrenewable Output as Percentage of Total Load

